

ACID ASSIGNMENT

Query 1 x SQL File 6*

Limit to 1000 rows

```
1 • use hr_acid_db;  
2 • select * from hr_acid_db.departments;
```

Result Grid | Filter Rows: | Edit: | Export/

	department_id	department_name	location
▶	1	HR	Yangon
	2	IT	Mandalay
	3	Finance	Naypyidaw
	4	Admin	Yangon
	5	Operations	Bago
*	NULL	NULL	NULL

Department table

Query 1 x SQL File 6*

Limit to 1000 rows

```
1 • use hr_acid_db;  
2 • select * from hr_acid_db.employees;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

	employee_id	employee_code	employee_name	department_id	job_title	salary	leave_balance	status
▶	1	EMP001	Aung Aung	2	Junior Developer	800000.00	10	Active
	2	EMP002	Su Su	1	HR Officer	650000.00	8	Active
	3	EMP003	Kyaw Kyaw	3	Accountant	900000.00	12	Active
	4	EMP004	Hla Hla	4	Admin Assistant	500000.00	7	Active
	5	EMP005	Mya Mya	2	System Engineer	1200000.00	15	Active
	6	EMP006	Ko Ko	5	Operations Officer	700000.00	9	Active
	7	EMP007	Nyein Nyein	1	Recruitment Specialist	680000.00	11	Active
	8	EMP008	Zaw Zaw	3	Finance Analyst	950000.00	14	Active
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Employees table

Query 1 x SQL File 6*

Limit to 1000 rows

```

1 • use hr_acid_db;
2 • select * from hr_acid_db.leave_requests;

```

Result Grid

	leave_request_id	employee_id	leave_type	days_requested	request_status	request_date	approved_by
1	1	1	Annual Leave	2	Approved	2026-04-01	HR Manager
2	2	2	Medical Leave	1	Approved	2026-04-03	HR Manager
3	3	4	Casual Leave	2	Pending	2026-04-05	NULL
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Leave_request table

Query 1 x SQL File 6*

Limit to 1000 rows

```

1 • use hr_acid_db;
2 • select * from hr_acid_db.payroll_transactions;

```

Result Grid

	payroll_id	employee_id	payroll_month	basic_salary	bonus	deduction	net_salary	processed_at
1	1	1	2026-04	800000.00	50000.00	10000.00	840000.00	2026-04-26 13:52:54
2	2	2	2026-04	650000.00	30000.00	5000.00	675000.00	2026-04-26 13:52:54
3	3	3	2026-04	900000.00	40000.00	15000.00	925000.00	2026-04-26 13:52:54
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Payroll_transactions table

ACID TEST

ACID TEST 1 – ATOMICITY

ATOMICITY TEST BEFORE SUCCESSFUL TRANSACTION

```
2
3 • SELECT 'ATOMICITY TEST - BEFORE' AS info;
4 • SELECT employee_id, employee_name, leave_balance
5 FROM employees
6 WHERE employee_id = 1;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Co
	employee_id	employee_name	leave_balance	
▶	1	Aung Aung	10	
*	NULL	NULL	NULL	

```
8 • SELECT * FROM leave_requests WHERE employee_id = 1;
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	leave_request_id	employee_id	leave_type	days_requested	request_status	request_date	approved_by
▶	1	1	Annual Leave	2	Approved	2026-04-01	HR Manager
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

ATOMICITY TEST : SUCCESSFUL TRANSACTION

-- 4A. Successful atomic transaction

START TRANSACTION;

INSERT INTO leave_requests (employee_id, leave_type, days_requested, request_status, request_date, approved_by) VALUES (1, 'Annual Leave', 3, 'Approved', CURDATE(), 'HR Manager');

UPDATE employees SET leave_balance = leave_balance - 3 WHERE employee_id = 1;

COMMIT;

```
167 • SELECT 'ATOMICITY TEST - AFTER SUCCESSFUL COMMIT' AS info;
168 • SELECT employee_id, employee_name, leave_balance
169 FROM employees
170 WHERE employee_id = 1;
171
```

employee_id	employee_name	leave_balance
1	Aung Aung	7
*	NULL	NULL

```
172 • SELECT * FROM leave_requests WHERE employee_id = 1;
173
```

leave_request_id	employee_id	leave_type	days_requested	request_status	request_date	approved_by
1	1	Annual Leave	2	Approved	2026-04-01	HR Manager
4	1	Annual Leave	3	Approved	2026-04-30	HR Manager
*	NULL	NULL	NULL	NULL	NULL	NULL

ATOMICITY TEST BEFORE FAILED TRANSACTION

179 • SELECT 'ATOMICITY TEST - BEFORE FAILED TRANSACTION' AS info;

180 • SELECT employee_id, employee_name, leave_balance

181 FROM employees

182 WHERE employee_id = 2;

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	employee_id	employee_name	leave_balance
▶	2	Su Su	8
✱	NULL	NULL	NULL

ATOMICITY TEST : FAILED TRANSACTION

-- We intentionally create an error by using invalid employee_id in leave_requests

-- and then rollback everything.

START TRANSACTION;

-- Step 1: Deduct leave balance

UPDATE employees

SET leave_balance = leave_balance - 2

WHERE employee_id = 2;

-- Step 2: Invalid insert to force error

-- employee_id 9999 does not exist, so FK constraint fails

-- Depending on MySQL client, execute this line and then run ROLLBACK manually

INSERT INTO leave_requests

(employee_id, leave_type, days_requested, request_status, request_date, approved_by)

VALUES

(9999, 'Annual Leave', 2, 'Approved', CURDATE(), 'HR Manager');

ROLLBACK;

```
200
201 • SELECT 'ATOMICITY TEST - AFTER ROLLBACK' AS info;
202 • SELECT employee_id, employee_name, leave_balance
203 FROM employees
204 WHERE employee_id = 2;
205
206 -- =====
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap
employee_id	employee_name	leave_balance		
2	Su Su	8		
* NULL	NULL	NULL		

ACID TEST 2 – CONSISTENCY

CONSISTENCY TEST BEFORE SUCCESSFUL TRANSACTION

```

217
218 • SELECT 'CONSISTENCY TEST - BEFORE' AS info;
219 • SELECT employee_id, employee_name, salary, leave_balance
220 FROM employees
221 WHERE employee_id = 3;

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Co
employee_id	employee_name	salary	leave_balance	
3	Kyaw Kyaw	900000.00	12	
NULL	NULL	NULL	NULL	*

Valid transaction

START TRANSACTION;

UPDATE employees

SET leave_balance = leave_balance - 5

WHERE employee_id = 3

AND leave_balance >= 5;

COMMIT;

CONSISTENCY TEST - AFTER VALID UPDATE

```

236 • SELECT 'CONSISTENCY TEST - AFTER VALID UPDATE' AS info;
237 • SELECT employee_id, employee_name, salary, leave_balance
238 FROM employees
239 WHERE employee_id = 3;

```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap C
employee_id	employee_name	salary	leave_balance	
3	Kyaw Kyaw	900000.00	7	
NULL	NULL	NULL	NULL	*

Invalid transaction example

START TRANSACTION;

UPDATE employees

SET leave_balance = -1

WHERE employee_id = 4;

ROLLBACK;

CONSISTENCY TEST - AFTER INVALID SALARY ROLLBACK

```
271 • SELECT 'CONSISTENCY TEST - AFTER INVALID SALARY ROLLBACK' AS info;  
272 • SELECT employee_id, employee_name, salary, leave_balance  
273 FROM employees  
274 WHERE employee_id = 5;
```

Result Grid	Filter Rows:	Edit:	Export/Import:	Wrap Cell Content
employee_id	employee_name	salary	leave_balance	
5	Mya Mya	1200000.00	15	
NULL	NULL	NULL	NULL	

ACID TEST 4 – DURABILITY TEST

DURABILITY TEST – BEFORE


```

333 • SELECT 'DURABILITY TEST - BEFORE' AS info;
334 • SELECT * FROM payroll_transactions WHERE employee_id = 7;
335

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	payroll_id	employee_id	payroll_month	basic_salary	bonus	deduction	net_salary	processed_at
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

INSERT INTO payroll_transactions

(employee_id, payroll_month, basic_salary, bonus, deduction, net_salary)

VALUES

(7, '2026-05', 680000, 50000, 10000, 720000);

COMMIT;

DURABILITY TEST - AFTER COMMIT

```

345 • SELECT 'DURABILITY TEST - AFTER COMMIT' AS info;
346 • SELECT * FROM payroll_transactions WHERE employee_id = 7;

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	payroll_id	employee_id	payroll_month	basic_salary	bonus	deduction	net_salary	processed_at
▶	4	7	2026-05	680000.00	50000.00	10000.00	720000.00	2026-04-30 12:11:27
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

FULL BUSINESS TEST - BEFORE PROMOTION

```

366
367 • SELECT 'FULL BUSINESS TEST - BEFORE PROMOTION' AS info;
368 • SELECT * FROM employees WHERE employee_id = 8;
369

```

employee_id	employee_code	employee_name	department_id	job_title	salary	leave_balance	status	hire_date
8	EMP008	Zaw Zaw	3	Finance Analyst	950000.00	14	Active	2021-05-01
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

START TRANSACTION;

UPDATE employees

SET department_id = 2,

job_title = 'Senior Finance Systems Analyst',

salary = salary + 150000

WHERE employee_id = 8;

INSERT INTO payroll_transactions

(employee_id, payroll_month, basic_salary, bonus, deduction, net_salary)

VALUES

(8, '2026-05', 1100000, 100000, 20000, 1180000);

COMMIT;

FULL BUSINESS TEST - AFTER PROMOTION COMMIT

```

385 • SELECT 'FULL BUSINESS TEST - AFTER PROMOTION COMMIT' AS info;
386 • SELECT * FROM employees WHERE employee_id = 8;

```

Result Grid Filter Rows: Edit: Export/Import: Wrap Cell Content:							
	employee_id	employee_code	employee_name	department_id	job_title	salary	leave_balance
▶	8	EMP008	Zaw Zaw	2	Senior Finance Systems Analyst	1100000.00	14
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL

```

385 • SELECT 'FULL BUSINESS TEST - AFTER PROMOTION COMMIT' AS info;
386 • SELECT * FROM employees WHERE employee_id = 8;
387 • SELECT * FROM payroll_transactions WHERE employee_id = 8;

```

Result Grid		Filter Rows:		Edit:	Export/Import:		Wrap Cell Content:	
	payroll_id	employee_id	payroll_month	basic_salary	bonus	deduction	net_salary	processed_at
	5	8	2026-05	1100000.00	100000.00	20000.00	1180000.00	2026-04-30 12:13:03
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Failed promotion transaction example

FULL BUSINESS TEST - BEFORE FAILED PROMOTION

```

393 • SELECT 'FULL BUSINESS TEST - BEFORE FAILED PROMOTION' AS info;
394 • SELECT * FROM employees WHERE employee_id = 5;
395

```

Result Grid										Filter Rows:		Edit:			Export/Import:		Wrap Cell Content:		Result Grid
	employee_id	employee_code	employee_name	department_id	job_title	salary	leave_balance	status	hire_date										
▶	5	EMP005	Mya Mya	2	System Engineer	1200000.00	15	Active	2026-04-30										
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL										

START TRANSACTION;

UPDATE employees

```

SET job_title = 'Lead System Engineer',
    salary = salary + 200000
WHERE employee_id = 5;
-- Invalid foreign key department_id = 9999
UPDATE employees
SET department_id = 9999
WHERE employee_id = 5;
ROLLBACK;

```

FULL BUSINESS TEST - AFTER FAILED PROMOTION ROLLBACK

```

410 • SELECT 'FULL BUSINESS TEST - AFTER FAILED PROMOTION ROLLBACK' AS info;
411 • SELECT * FROM employees WHERE employee_id = 5;
...

```

	employee_id	employee_code	employee_name	department_id	job_title	salary	leave_balance	status	hire_date
▶	5	EMP005	Mya Mya	2	System Engineer	1200000.00	15	Active	2023-01-01
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

FINAL DATA CHECK

FINAL EMPLOYEE DATA

```

417 • SELECT
418     e.employee_id,
419     e.employee_code,
420     e.employee_name,
421     d.department_name,
422     e.job_title,
423     e.salary,
424     e.leave_balance,
425     e.status,
426     e.hire_date
427 FROM employees e
428 LEFT JOIN departments d
429     ON e.department_id = d.department_id
430 ORDER BY e.employee_id;
431

```

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	employee_id	employee_code	employee_name	department_name	job_title	salary	leave_balanc
▶	1	EMP001	Aung Aung	IT	Junior Developer	800000.00	7
	2	EMP002	Su Su	HR	HR Officer	650000.00	8
	3	EMP003	Kyaw Kyaw	Finance	Accountant	900000.00	7
	4	EMP004	Hla Hla	Admin	Admin Assistant	500000.00	7
	5	EMP005	Mya Mya	IT	System Engineer	1200000.00	15
	6	EMP006	Ko Ko	Operations	Operations Officer	700000.00	9
	7	EMP007	Nyein Nyein	HR	Recruitment Specialist	680000.00	11
	8	EMP008	Zaw Zaw	IT	Senior Finance Systems Analyst	1100000.00	14

FINAL PAYROLL DATA

```

432 • SELECT 'FINAL PAYROLL DATA' AS info;
433 • SELECT * FROM payroll_transactions ORDER BY payroll_id;

```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	payroll_id	employee_id	payroll_month	basic_salary	bonus	deduction	net_salary	processed_at
▶	1	1	2026-04	800000.00	50000.00	10000.00	840000.00	2026-04-26 13:52:54
	2	2	2026-04	650000.00	30000.00	5000.00	675000.00	2026-04-26 13:52:54
	3	3	2026-04	900000.00	40000.00	15000.00	925000.00	2026-04-26 13:52:54
	4	7	2026-05	680000.00	50000.00	10000.00	720000.00	2026-04-30 12:11:27
	5	8	2026-05	1100000.00	100000.00	20000.00	1180000.00	2026-04-30 12:13:03

FINAL LEAVE REQUEST DATA

```
435 • SELECT 'FINAL LEAVE REQUEST DATA' AS info;
```

```
436 • SELECT * FROM leave_requests ORDER BY leave_request_id;
```

[illegible]